Bernoulli recurrences for the quintic holomorphic ambiguity

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The proposed Bernoulli structure does extend into the low powers left unfixed by the conifold gap. There is an exact convolution for the coefficient of X, a small exact recursion for any fixed Taylor order with the requisite MUM data, and a useful formal large-genus approximation for further coefficients. The accompanying computation obtains the coefficients through X^4 exactly at every genus through 53. Full polynomials have been computed through genus 12 only. These are distinct computational claims.

1 Conventions and the exact low-power formula

Use the same zero-at-$v = 0$ integration convention as the earlier scan:

$$P_g = P_g^{\mathrm{an}} + a_g(X), \qquad P_g^{\mathrm{an}}(v_1 = v_2 = v_3 = 0, X) = 0, \qquad a_g(X) = \sum_{n=0}^{3g-3} a_{g,n} X^n.$$

Here $x = z/\mu = \psi^{-5}$ and $X = x/(x-1)$. The rational ring coefficients are independent of the coordinate scale μ. Write

$$b_g = |B_{2g}|, \qquad c_g := a_{g,0} = 100\,625^{g-1} \frac{b_g b_{g-1}}{2g(2g-2)(2g-2)!}, \qquad g \geq 2. \tag{1}$$

All these numbers can be generated from

$$b_1 = \frac{1}{6}, \qquad b_g = \frac{1}{2g+1} \sum_{h=1}^{g-1} \binom{2g}{2h} b_h b_{g-h}.$$

This is the classical Euler Bernoulli convolution with signs removed [2]. Define

$$S_g = \sum_{h=2}^{g-2} (2h-2)(2g-2h-2) c_h c_{g-h}, \tag{2}$$

$$I_g = \frac{23}{25} (-625)^{g-1} \frac{b_g}{2g(2g-2)!}. \tag{3}$$

Then the derived identity is

$$\boxed{\begin{aligned} a_{g,1} = & -\frac{577}{625}(g-1)c_g + (2g-4)\left(62 - \frac{24g}{5}\right) c_{g-1} \\ & - \frac{12}{5} S_g - I_g, \qquad g \geq 3. \end{aligned}} \tag{4}$$

An empty sum is zero. Genus two is the separate seed $a_{2,1} = -28795/144$. This identity was derived from the BCOV and MUM equations, without sequence fitting. It agrees with the full physical polynomials through genus 12 and the independent Taylor computation through genus 53. The term I_g contains the quintic degree-one invariant 2875; Bernoulli numbers alone, without geometric input, do not fix an arbitrary model.

2 Derivation of the convolution

Let $P_h^{(r)}$ denote a covariantly differentiated YY vertex. At $v_1 = v_2 = v_3 = X = 0$, while retaining u, the differential ring gives

$$\theta u = -u^2, \qquad \theta v_i = 0, \qquad P_h^{(1)} = (2h-2)uc_h, \qquad P_h^{(2)} = (2h-2)(2h-1)u^2 c_h \quad (h \geq 2).$$

The genus-one seed is $P_1^{(1)} = 125/12 - (28/3)u$ on this locus. Insert these expressions into

$$R_g = -\frac{1}{2}\left(P_{g-1}^{(2)} + \sum_{h=1}^{g-1} P_h^{(1)} P_{g-h}^{(1)}\right), \qquad (\partial_{v_1} - u\partial_{v_2} - u(u+5X)\partial_{v_3})P_g = R_g.$$

Writing $m_i = [v_i X^0]P_g$ gives, for $g \geq 3$,

$$\begin{aligned} m_1 &= 0, \\ m_2 &= \frac{125}{12}(2g-4)c_{g-1}, \\ m_3 &= \frac{1}{2}\left[(2g-4)\left(2g - \frac{65}{3}\right)c_{g-1} + S_g\right]. \end{aligned}$$

At the MUM point, the needed period data are just

$$X = -x + O(x^2), \quad v_1 = -\frac{202}{125}x + O(x^2), \quad v_2 = \frac{24}{25}x + O(x^2), \quad v_3 = -\frac{24}{5}x + O(x^2),$$
$$k(x) = \frac{f(x)^2(x-1)}{625} = -\frac{1}{625}\left(1 - \frac{577}{625}x + O(x^2)\right), \qquad q = \frac{x}{3125} + O(x^2),$$

where the physical amplitude is $F_g^{\mathrm{MUM}} = k^{g-1}P_g^{\mathrm{hol}}$. The degree-one GW coefficient is $2875\, b_g/[2g(2g-2)!]$: the positive-genus degree-one GV invariants vanish. Equating the coefficients of x in the MUM amplitude therefore gives

$$a_{g,1} = -\frac{577}{625}(g-1)c_g + \frac{24}{25}m_2 - \frac{24}{5}m_3 - I_g,$$

which reduces to (4).

3 A short approximation with measured accuracy

The factorial growth makes the convolution (2) concentrate at its ends. For $g \geq 5$, retaining only $h = 2$ and $h = g-2$ gives

$$S_g^{(2)} = 8(g-3)c_2 c_{g-2}.$$

Substitute this for S_g in (4), retaining the exact I_g. There are no fitted parameters. Relative errors, evaluated against the exact recurrence, are

Genus	12	16	30	53
Error (%)	0.0597094	0.00647730	0.0000961011	0.00000274661

Additional endpoint pairs systematically improve this finite-genus test. The script handles the middle term without double counting.

The underlying scale can also be seen from $b_g = 2(2g)!\zeta(2g)/(2\pi)^{2g}$ [2]. In particular,

$$\frac{c_{g-1}}{c_g} \sim \frac{16\pi^4}{625(2g-1)(2g-4)}.$$

Keeping the first endpoint pair produces the formal expansion

$$\frac{a_{g,1}}{c_g} \sim -\frac{577}{625}(g-1) - \frac{192\pi^4}{3125} + \frac{4768\pi^4}{3125(2g-1)} - \frac{256\pi^8}{1875(2g-1)(2g-3)(2g-4)} + O(g^{-5}).$$

The displayed rational functions retain more than a finite $1/g$ expansion; there are also exponentially small zeta corrections and a still smaller alternating degree-one curve contribution. For computation at modest genus, the exact Bernoulli-ratio version above is preferable.

4 Exact low-power recursion without full polynomials

Define the ideal $J = (v_1, v_2, v_3, X)$; u is deliberately excluded from its degree. Every θv_i and θX vanishes on $J = 0$, even at arbitrary u. Thus θ preserves every power of J. Products and extraction of u coefficients preserve this filtration too. Integration in v_i raises the J degree by one.

To determine an amplitude through J degree N, it therefore suffices to compute its anomaly gradient through degree $N-1$. After integration, impose the MUM amplitude through x^N. All MUM generators, including u, vanish at $x = 0$, so discarded terms cannot contribute to the retained Taylor coefficients. No estimate or floating-point approximation enters this step. Pure X^n boundary coefficients are fixed by an ordinary triangular Taylor matching, since $X = -x + O(x^2)$.

This also explains the recurrence structure: the anomaly part relevant to $a_{g,n}$ only needs lower-genus Taylor data of J degree at most $n-1$. The required MUM coefficients are known constant maps and low-degree GV multicover contributions. Once Castelnuovo vanishing applies at the retained curve degrees, there is no new same-genus enumerative input at those degrees. The supplied seed checkpoint provides the remaining finite low-genus data.

The computation with $N = 4$ retains 66 nonzero amplitude monomials per genus from genus five onward. It obtains $a_{g,0}, \ldots, a_{g,4}$ through genus 53 in exact rationals, instead of expanding the full degree-$3g-3$ polynomials. At genus two, $a_{2,4} = 0$ by the degree bound. Checks include all retained anomaly terms against the full checkpoint through genus 12, 54 stored pure coefficients against those full amplitudes, and the first three pure coefficients against a separate X-only quotient calculation through genus 16. The latter retains many more v monomials and agrees exactly. The explicit formula (4), including its genus-two seed, passes all 52 comparisons through genus 53.

5 A Bernoulli/Bell approximation for further coefficients

Let $r_{g,n} = a_{g,n}/c_g$. It is useful to organize their logarithm,

$$\log\left(\frac{a_g(X)}{c_g}\right) = \sum_{j\geq 1} \ell_{g,j} X^j.$$

Products of the large linear coefficient otherwise hide the simpler structure. The following is a formal large-genus approximation at *fixed* Taylor order, not an approximation uniform in n or throughout moduli space.

Express the MUM periods as functions of X, using $x = X/(X-1)$, and define

$$H(X) = k(x(X))/k(0), \qquad L(X) = -\log H(X),$$
$$T(X) = 5X(X-1)L'(X) - 5X, \qquad \gamma = \frac{4\pi^4}{625},$$
$$S(X) = \frac{H(X)}{2}\left[T^2 v_1^{\mathrm{MUM}} - 4(T-5X)v_2^{\mathrm{MUM}} - 4v_3^{\mathrm{MUM}}\right].$$

The formal leading BCOV balance predicts

$$\log\left(a_g(X)/c_g\right) \sim (g-1)L(X) + \gamma S(X) + O(g^{-1}) \tag{5}$$

coefficientwise in X. To see its origin, use $c_{g-1}/c_g \sim \gamma/g^2$. The handle term $P_{g-1}^{(2)}$ in the anomaly is then leading, while genus-splitting endpoint terms start at lower powers of $1/g$. The leading one-point multiplier is $g(2u+T)$, so the anomaly source divided by P_g tends to $-\gamma H(2u+T)^2/2$. Integrating its three u coefficients in the v_i and imposing the MUM value gives precisely $\gamma S(X)$ at $v=0$. This is a formal asymptotic derivation, not a proof of a uniform error bound.

For example,

$$L(X) = -\frac{577}{625}X - \frac{347617}{781250}X^2 + O(X^3),$$
$$S(X) = -\frac{48}{5}X - \frac{29424}{3125}X^2 + O(X^3).$$

The first term reproduces the leading behavior of the exact formula for $a_{g,1}/c_g$. For better finite-genus accuracy, keep that linear coefficient *exact*, and approximate only the others:

$$\widehat{\ell}_{g,1} = a_{g,1}/c_g, \qquad \widehat{\ell}_{g,j} = (g-1)[X^j]L + \gamma[X^j]S \quad (j \geq 2).$$

The coefficients are generated by the Bell-polynomial recurrence

$$\boxed{\widehat{r}_{g,0} = 1, \qquad n\widehat{r}_{g,n} = \sum_{j=1}^{n} j\widehat{\ell}_{g,j}\widehat{r}_{g,n-j}.} \tag{6}$$

No coefficient was fitted to the genus data. This model uses the exact Bernoulli convolution only for $r_{g,1}$; $r_{g,2}, r_{g,3}, r_{g,4}$ are predictions. Measured relative errors against the exact Taylor computation:

g	X^2 error (%)	X^3 error (%)	X^4 error (%)
12	13.7775	46.3902	97.2279
16	3.96512	13.1014	28.4254
20	1.77043	5.72480	12.2735
30	0.486136	1.52730	3.19581
40	0.206381	0.639896	1.32237
53	0.090804	0.279004	0.571497

It is useful at larger genus and can be poor at small genus. The existing exact Taylor recursion is both more accurate and inexpensive for these fixed low powers; the asymptotic model exposes their structure.

6 Reproduction and limits

Run from the extracted directory with Python 3.10 or newer:

```
python compute_low_x.py --genus 53 --degree 4 --jet \
  --output low_x_degree4_through53.json
python analyze_recurrence.py
```

Both commands use only the standard library. The optional figure script uses Matplotlib. The full physical checkpoint, exact results, independent quotient results, source modules and verification metadata are included. The original physical runner supplied the fresh full genera 11 and 12; its per-genus boundary diagnostics accompany that checkpoint.

The results establish useful Bernoulli-driven recurrences in the stated integration convention. They do not identify a universal Bernoulli sequence that fixes arbitrary holomorphic ambiguities without geometric boundary data. They also do not yet control the middle coefficients with n increasing proportionally to g, or replace the conifold and monodromy constraints needed for the full polynomial. Exact rational numerator growth remains, but the low-degree computation avoids the much larger polynomial expansion.

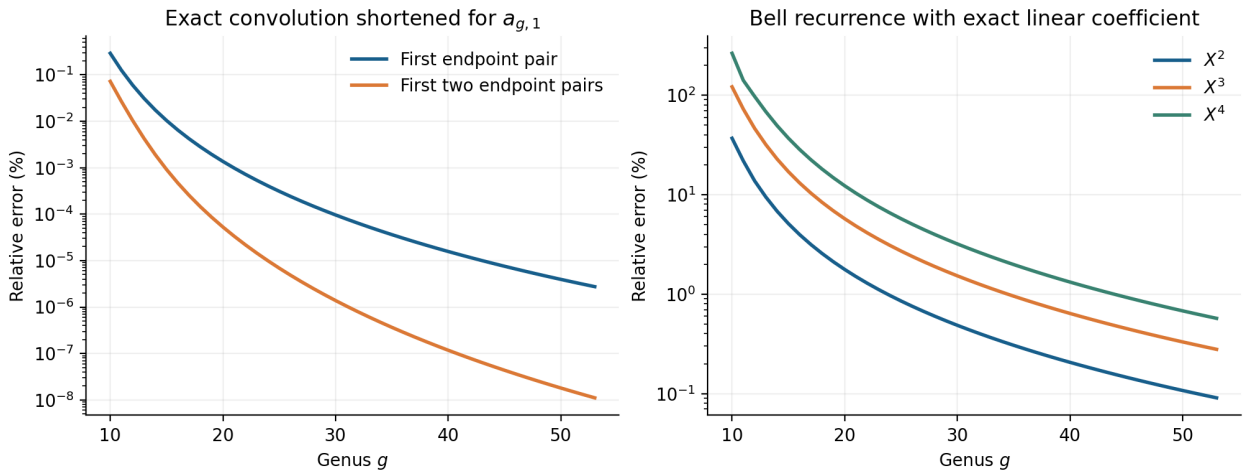


Figure 1: Measured errors through genus 53. Left: truncate the exact Bernoulli convolution. Right: use (6), fixing only the linear logarithmic coefficient exactly. Values above genus 12 are tested against exact Taylor coefficients, not full amplitudes.

References

[1] S. Yamaguchi and S.-T. Yau, *Topological String Partition Functions as Polynomials*, hep-th/0406078.

[2] G. V. Dunne and C. Schubert, *Bernoulli Number Identities from Quantum Field Theory and Topological String Theory*, math/0406610v2. This supplies the classical Euler and related Bernoulli convolution identities and the Bernoulli–zeta formula, not the quintic-specific identity derived here.